$$\begin{tikzcd}
F(A) \arrow[r, "F(f)"] \arrow[d, "\eta_A"'] & F(B) \arrow[d, "\eta_B"] \\
G(A) \arrow[r, "G(f)"'] & G(B)
\end{tikzcd}$$

Figure 1: Natural Transformation: $\eta : F \to G$

$$G(\text{🦇}) \xrightarrow{n=\langle \text{💀}, \text{👻} \rangle} \text{👻}\ \text{🎃}_{i=1}^{n} H_i(\text{🧙})$$

Figure 2: Happy Halloween!

$$A \xrightarrow[abc\ldots z]{f_1+\cdots+f_n} B \xrightarrow{f_1+\cdots+f_n} C \xrightarrow[abc\ldots z]{} D$$

Figure 3: Happy Halloween!

🧙🎃💀☁️ ➢ 👻🦇

0.6249999121401877

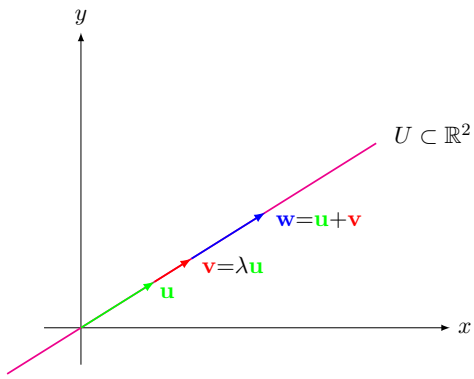

Figure 4: The line U is a linear subspace of $\mathbb{R}^2$

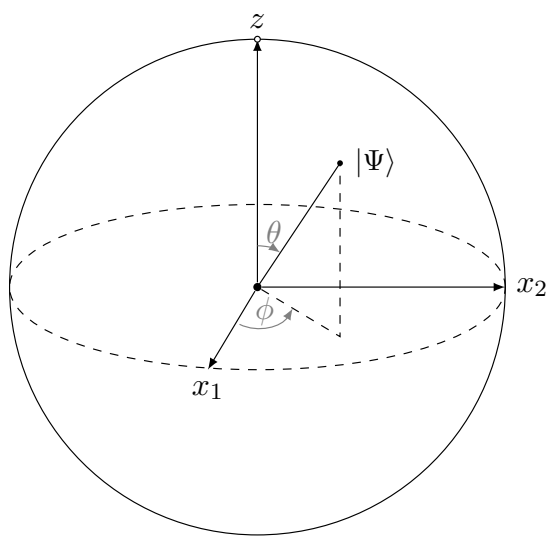

Figure 5: The Bloch Sphere

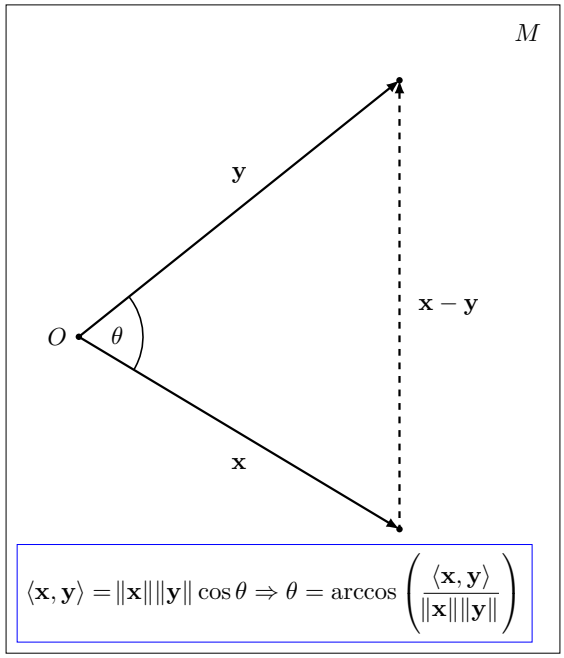

Figure 6: Geometric interpretation of $\langle \mathbf{x}, \mathbf{y} \rangle$

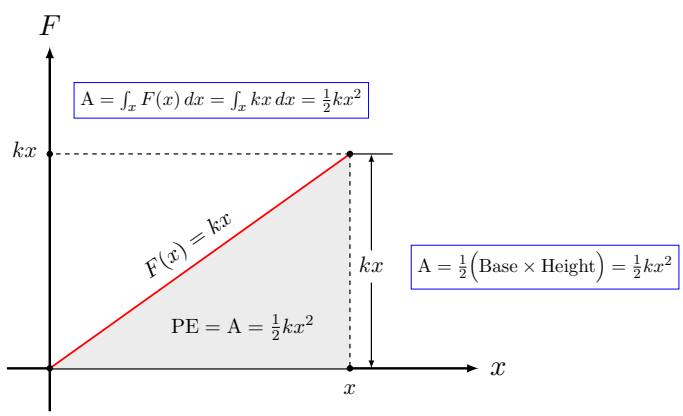

Figure 7: Potential Energy of a Simple Harmonic Oscillator

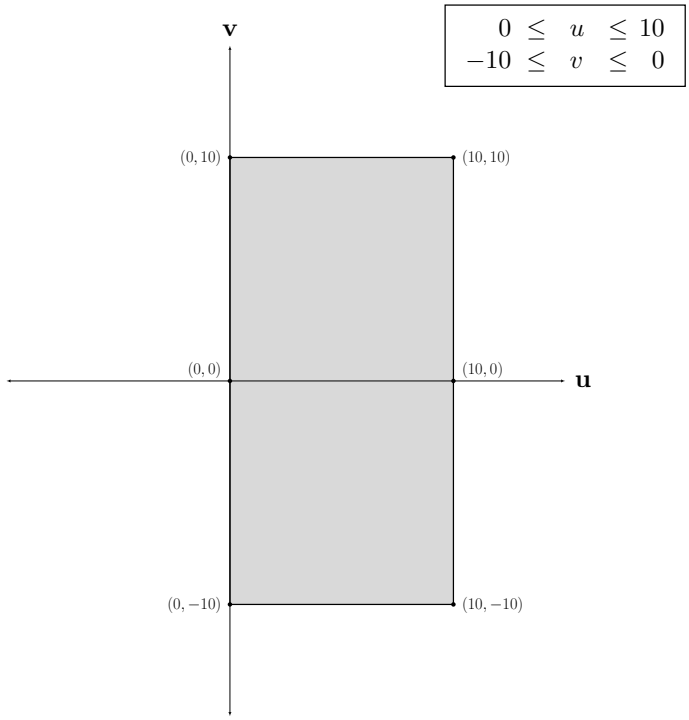

Figure 8: The transformed parallelogram is a rectangle in the uv-plane

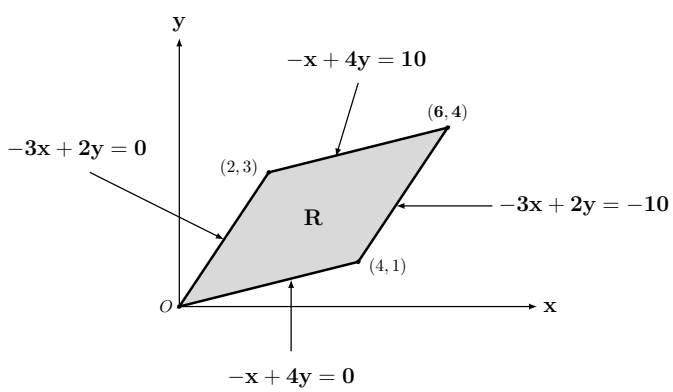

Figure 9: Equations of the Sides of the Parallelogram

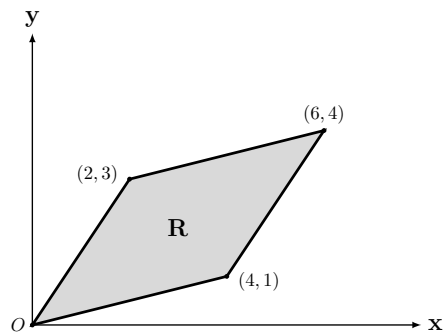

Figure 10: Parallelogram in the xy-plane

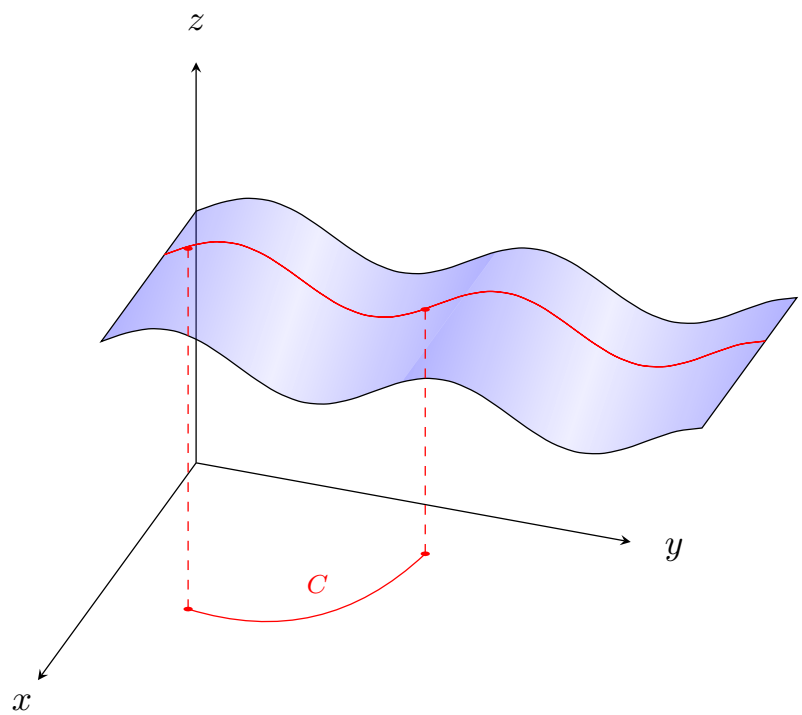

Figure 11: 3D

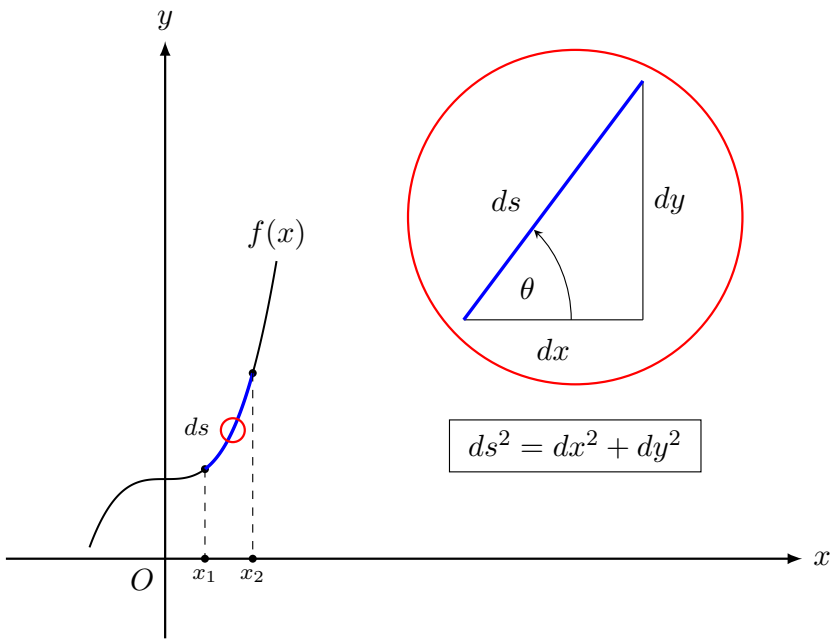

Figure 12: $f(x)$, ds and the Pythagorean Theorem

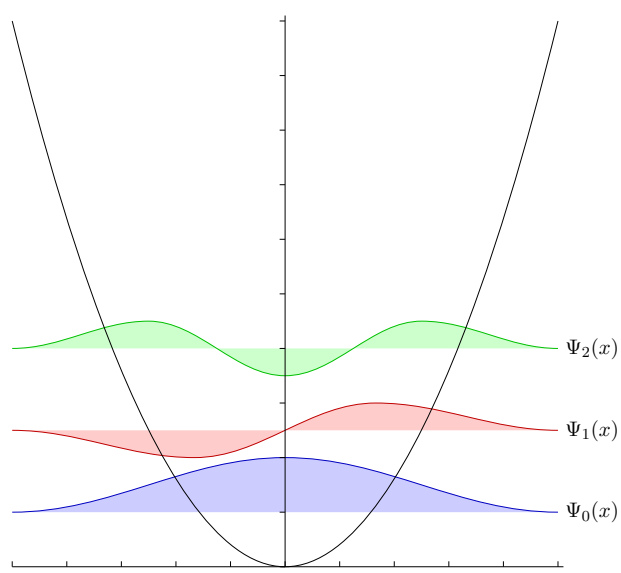

Figure 13: Wave Functions

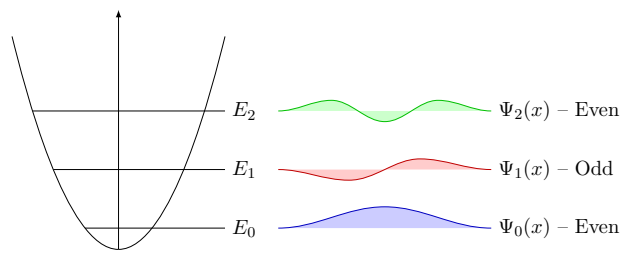

Figure 14: Wave Functions

9^2	$=$	81	$\longrightarrow$	$8+1$	$=$	9
45^2	$=$	2025	$\longrightarrow$	$20+25$	$=$	45
703^2	$=$	494209	$\longrightarrow$	$494+209$	$=$	703
7777^2	$=$	60481729	$\longrightarrow$	$6048+1729$	$=$	7777
857143^2	$=$	734694122449	$\longrightarrow$	$734694+122449$	$=$	857143

Figure 15: A Few Example Kaprekar Numbers

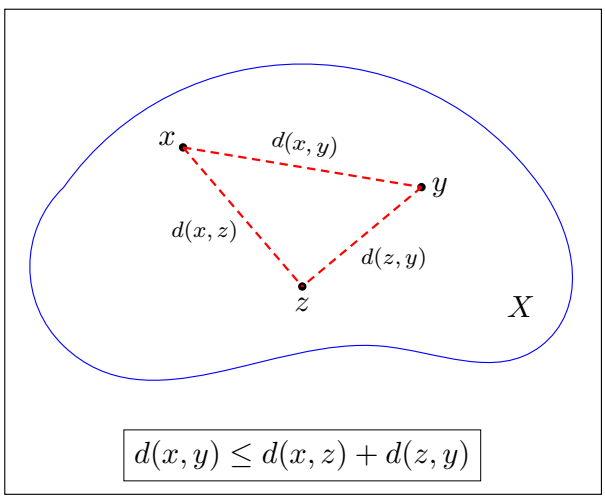

Figure 16: The Triangle Inequality

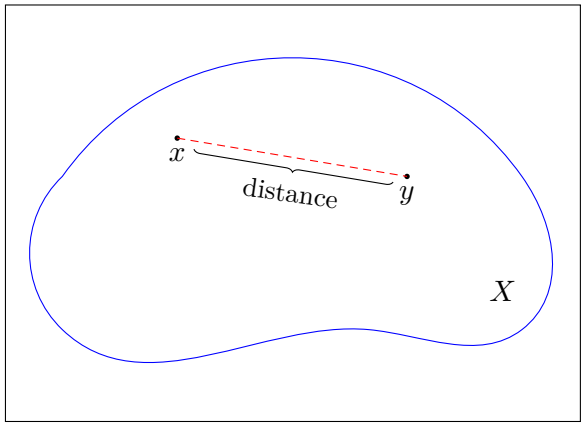

Figure 17: X, x and y

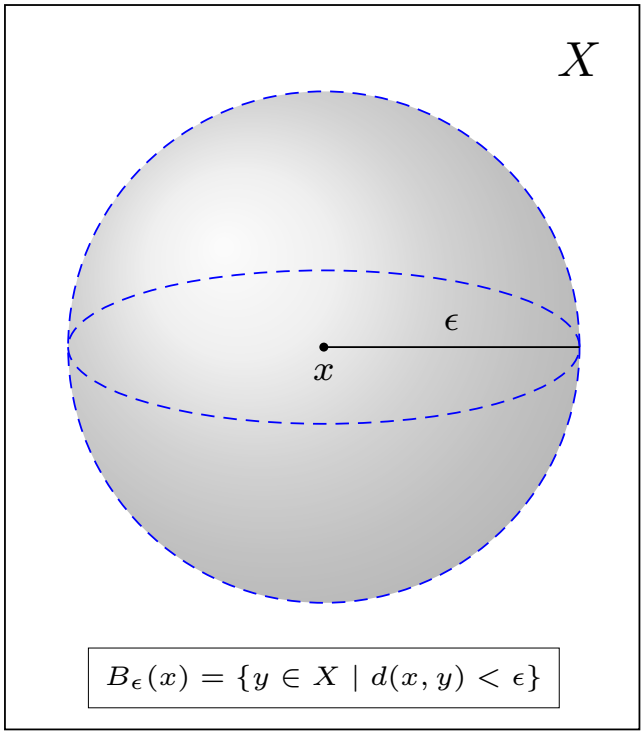

Figure 18: $B_\epsilon(x)$ is an open epsilon ball centered at x with radius ϵ

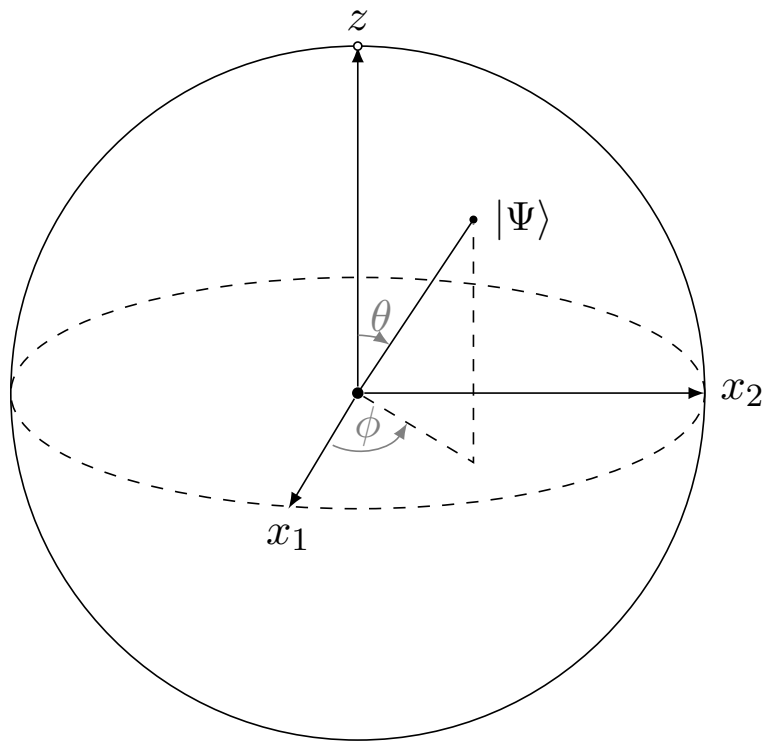

Figure 19: The Bloch Sphere

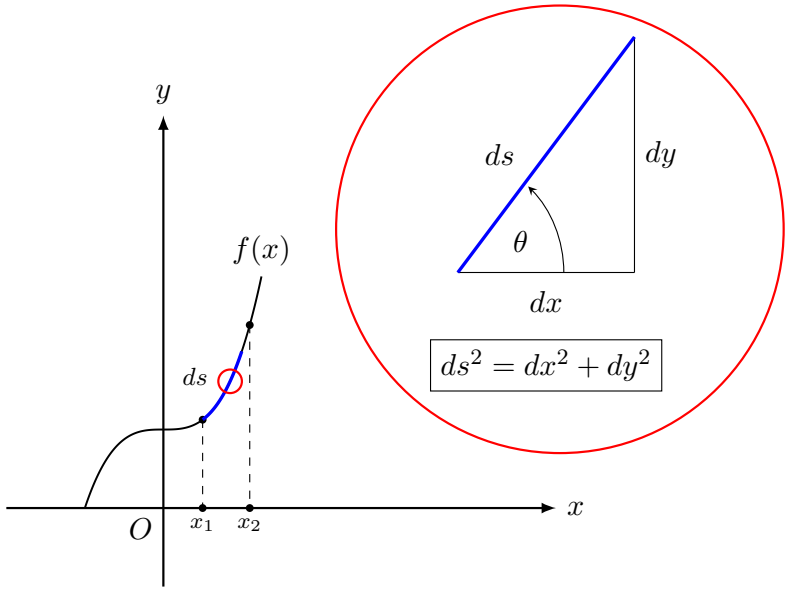

Figure 20: $f(x)$, ds and the Pythagorean Theorem

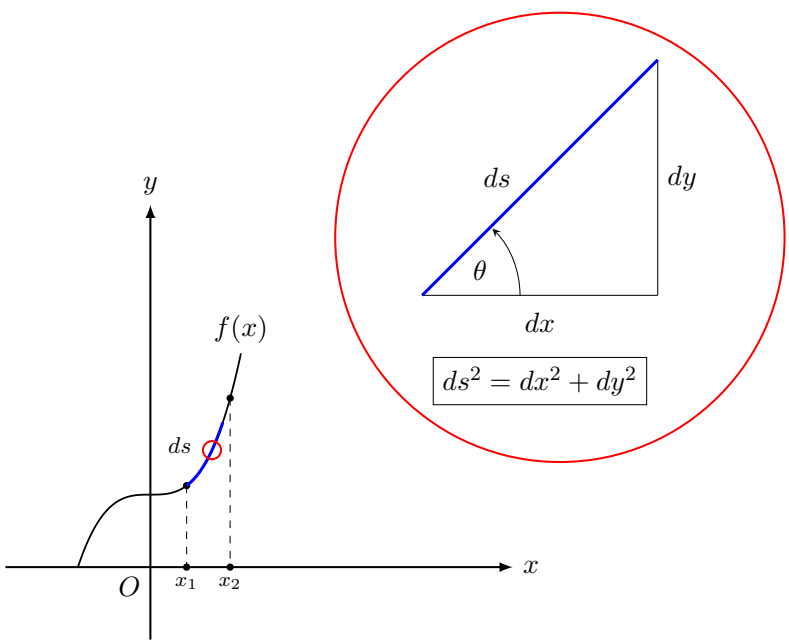

Figure 21: $f(x)$, ds and the Pythagorean Theorem